

1. Store the contents of PCL, PCH, X, A, and CCR on the stack, in that order.
2. Set the I bit in the CCR.
3. Fetch the high-order half of the interrupt vector.
4. Fetch the low-order half of the interrupt vector.
5. Delay for one free bus cycle.

Fetch three bytes of program information starting at the address indicated by the interrupt vector to fill the instruction queue in preparation for execution of the first instruction in the interrupt service routine.

After the CCR contents are pushed onto the stack, the I bit in the CCR is set to prevent other interrupts while in the interrupt service routine. Although it is possible to clear the I bit with an instruction in the interrupt service routine, this would allow nesting of interrupts (which is not recommended because it leads to programs that are difficult to debug and maintain).

For compatibility with the earlier M68HC05 MCUs, the high-order half of the H:X index register pair (H) is not saved on the stack as part of the interrupt sequence. The user must use a PSHH instruction at the beginning of the service routine to save H and then use a PULH instruction just before the RTI that ends the interrupt service routine. It is not necessary to save H if you are certain that the interrupt service routine does not use any instructions or auto-increment addressing modes that might change the value of H.

The software interrupt (SWI) instruction is like a hardware interrupt except that it is not masked by the global I bit in the CCR and it is associated with an instruction opcode within the program so it is not asynchronous to program execution.

10.7 Instruction Set Summary

Table 10-3. Instruction Set Summary

Source Form	Operation	Description	Effect on CCR						Address Mode	Opcode	Operand	Bus Cycles
			V	H	I	N	Z	C				
ADC #opr8i	Add with Carry	$A \leftarrow (A) + (M) + (C)$	↑	↑	—	↑	↑	↑	IMM	A9	ii	2
ADC opr8a			↑	↑	—	↑	↑	↑	DIR	B9	dd	3
ADC opr16a			↑	↑	—	↑	↑	↑	EXT	C9	hh ll	4
ADC oprx16,X			↑	↑	—	↑	↑	↑	IX2	D9	ee ff	4
ADC oprx8,X			↑	↑	—	↑	↑	↑	IX1	E9	ff	3
ADC ,X			↑	↑	—	↑	↑	↑	IX	F9		3
ADC oprx16,SP			↑	↑	—	↑	↑	↑	SP2	9ED9	ee ff	5
ADC oprx8,SP			↑	↑	—	↑	↑	↑	SP1	9EE9	ff	4

Table continues on the next page...

Table 10-3. Instruction Set Summary (continued)

Source Form	Operation	Description	Effect on CCR						Address Mode	Opcode	Operand	Bus Cycles
			V	H	I	N	Z	C				
ADD #opr8i	Add without Carry	$A \leftarrow (A) + (M)$	↑	↑	—	↑	↑	↑	IMM	AB	ii	2
ADD opr8a			↑	↑	—	↑	↑	↑	DIR	BB	dd	3
ADD opr16a			↑	↑	—	↑	↑	↑	EXT	CB	hh ll	4
ADD oprx16,X			↑	↑	—	↑	↑	↑	IX2	DB	ee ff	4
ADD oprx8,X			↑	↑	—	↑	↑	↑	IX1	EB	ff	3
ADD ,X			↑	↑	—	↑	↑	↑	IX	FB		3
ADD oprx16,SP			↑	↑	—	↑	↑	↑	SP2	9EDB	ee ff	5
ADD oprx8,SP			↑	↑	—	↑	↑	↑	SP1	9EEB	ff	4
AIS #opr8i	Add Immediate Value (Signed) to Stack Pointer	$SP \leftarrow (SP) + (M)$ where M is sign extended to a 16-bit value	—	—	—	—	—	—	IMM	A7	ii	2
AIX #opr8i	Add Immediate Value (Signed) to Index Register (H:X)	$H:X \leftarrow (H:X) + (M)$ where M is sign extended to a 16-bit value	—	—	—	—	—	—	IMM	AF	ii	2
AND #opr8i	Logical AND	$A \leftarrow (A) \& (M)$	0	—	—	↑	↑	—	IMM	A4	ii	2
AND opr8a			0	—	—	↑	↑	—	DIR	B4	dd	3
AND opr16a			0	—	—	↑	↑	—	EXT	C4	hh ll	4
AND oprx16,X			0	—	—	↑	↑	—	IX2	D4	ee ff	4
AND oprx8,X			0	—	—	↑	↑	—	IX1	E4	ff	3
AND ,X			0	—	—	↑	↑	—	IX	F4		3
AND oprx16,SP			0	—	—	↑	↑	—	SP2	9ED4	ee ff	5
AND oprx8,SP			0	—	—	↑	↑	—	SP1	9EE4	ff	4
ASL opr8a	Arithmetic Shift Left (same as LSL)	$C \leftarrow \text{MSB}, \text{LSB} \leftarrow 0$	↑	—	—	↑	↑	↑	DIR	38	dd	5
ASLA			↑	—	—	↑	↑	↑	INH	48		1
ASLX			↑	—	—	↑	↑	↑	INH	58		1
ASL oprx8,X			↑	—	—	↑	↑	↑	IX1	68	ff	5
ASL ,X			↑	—	—	↑	↑	↑	IX	78		4
ASL oprx8,SP			↑	—	—	↑	↑	↑	SP1	9E68	ff	6
ASR opr8a	Arithmetic Shift Right	$\text{MSB} \rightarrow \text{MSB}, \text{LSB} \rightarrow C$	↑	—	—	↑	↑	↑	DIR	37	dd	5
ASRA			↑	—	—	↑	↑	↑	INH	47		1
ASRX			↑	—	—	↑	↑	↑	INH	57		1
ASR oprx8,X			↑	—	—	↑	↑	↑	IX1	67	ff	5
ASR ,X			↑	—	—	↑	↑	↑	IX	77		4
ASR oprx8,SP			↑	—	—	↑	↑	↑	SP1	9E67	ff	6
BCC rel	Branch if Carry Bit Clear	Branch if (C) = 0	—	—	—	—	—	—	REL	24	rr	3
			—	—	—	—	—	—	DIR (b0)	11	dd	5

Table continues on the next page...

Table 10-3. Instruction Set Summary (continued)

Source Form	Operation	Description	Effect on CCR						Address Mode	Opcode	Operand	Bus Cycles
			V	H	I	N	Z	C				
BCLR n,opr8a	Clear Bit n in Memory	$M_n \leftarrow 0$	–	–	–	–	–	–	DIR (b1)	13	dd	5
			–	–	–	–	–	–	DIR (b2)	15	dd	5
			–	–	–	–	–	–	DIR (b3)	17	dd	5
			–	–	–	–	–	–	DIR (b4)	19	dd	5
			–	–	–	–	–	–	DIR (b5)	1B	dd	
			–	–	–	–	–	–	DIR (b6)	1D	dd	5
			–	–	–	–	–	–	DIR (b7)	1F	dd	5
BCS rel	Branch if Carry Bit Set (same as BLO)	Branch if (C) = 1	–	–	–	–	–	–	REL	25	rr	3
BEQ rel	Branch if Equal	Branch if (Z) = 1	–	–	–	–	–	–	REL	27	rr	3
BGE rel	Branch if Greater Than or Equal To (Signed Operands)	Branch if $(N \oplus V) = 0$	–	–	–	–	–	–	REL	90	rr	3
BGND	Enter Active Background if ENBDM = 1	Waits For and Processes BDM Commands Until GO, TRACE1, or TAGGO	–	–	–	–	–	–	INH	82		5+
BGT rel	Branch if Greater Than (Signed Operands)	Branch if $(Z) \mid (N \oplus V) = 0$	–	–	–	–	–	–	REL	92	rr	3
BHCC rel	Branch if Half Carry Bit Clear	Branch if (H) = 0	–	–	–	–	–	–	REL	28	rr	3
BHCS rel	Branch if Half Carry Bit Set	Branch if (H) = 1	–	–	–	–	–	–	REL	29	rr	3
BHI rel	Branch if Higher	Branch if $(C) \mid (Z) = 0$	–	–	–	–	–	–	REL	22	rr	3
BHS rel	Branch if Higher or Same (same as BCC)	Branch if (C) = 0	–	–	–	–	–	–	REL	24	rr	3
BIH rel	Branch if IRQ Pin High	Branch if IRQ pin = 1	–	–	–	–	–	–	REL	2F	rr	3
BIL rel	Branch if IRQ Pin Low	Branch if IRQ pin = 0	–	–	–	–	–	–	REL	2E	rr	3
BIT #opr8i	Bit Test	(A) & (M), (CCR Updated but Operands Not Changed)	0	–	–	↑	↑	–	IMM	A5	ii	2
BIT opr8a			0	–	–	↑	↑	–	DIR	B5	dd	3
BIT opr16a			0	–	–	↑	↑	–	EXT	C5	hh ll	4
BIT opr16,X			0	–	–	↑	↑	–	IX2	D5	ee ff	4
BIT opr8,X			0	–	–	↑	↑	–	IX1	E5	ff	3
BIT ,X			0	–	–	↑	↑	–	IX	F5		3
BIT opr16,SP			0	–	–	↑	↑	–	SP2	9ED5	ee ff	5
BIT opr8,SP			0	–	–	↑	↑	–	SP1	9EE5	ff	4

Table continues on the next page...

Table 10-3. Instruction Set Summary (continued)

Source Form	Operation	Description	Effect on CCR						Address Mode	Opcode	Operand	Bus Cycles
			V	H	I	N	Z	C				
BLE rel	Branch if Less Than or Equal To (Signed Operands)	Branch if $(Z) \vee (N \oplus V) = 1$	-	-	-	-	-	-	REL	93	rr	3
BLO rel	Branch if Lower (Same as BCS)	Branch if $(C) = 1$	-	-	-	-	-	-	REL	25	rr	3
BLS rel	Branch if Lower or Same	Branch if $(C) \vee (Z) = 1$	-	-	-	-	-	-	REL	23	rr	3
BLT rel	Branch if Less Than (Signed Operands)	Branch if $(N \oplus V) = 1$	-	-	-	-	-	-	REL	91	rr	3
BMC rel	Branch if Interrupt Mask Clear	Branch if $(I) = 0$	-	-	-	-	-	-	REL	2C	rr	3
BMI rel	Branch if Minus	Branch if $(N) = 1$	-	-	-	-	-	-	REL	2B	rr	3
BMS rel	Branch if Interrupt Mask Set	Branch if $(I) = 1$	-	-	-	-	-	-	REL	2D	rr	3
BNE rel	Branch if Not Equal	Branch if $(Z) = 0$	-	-	-	-	-	-	REL	26	rr	3
BPL rel	Branch if Plus	Branch if $(N) = 0$	-	-	-	-	-	-	REL	2A	rr	3
BRA rel	Branch Always	No Test	-	-	-	-	-	-	REL	20	rr	3
BRCLR n,opr8a,rel	Branch if Bit n in Memory Clear	Branch if $(Mn) = 0$	-	-	-	-	-	↕	DIR (b0)	01	dd rr	5
			-	-	-	-	-	↕	DIR (b1)	03	dd rr	5
			-	-	-	-	-	↕	DIR (b2)	05	dd rr	5
			-	-	-	-	-	↕	DIR (b3)	07	dd rr	5
			-	-	-	-	-	↕	DIR (b4)	09	dd rr	5
			-	-	-	-	-	↕	DIR (b5)	0B	dd rr	5
			-	-	-	-	-	↕	DIR (b6)	0D	dd rr	5
			-	-	-	-	-	↕	DIR (b7)	0F	dd rr	5
BRN rel	Branch Never	Uses 3 Bus Cycles	-	-	-	-	-	-	REL	21	rr	3
BRSET n,opr8a,rel	Branch if Bit n in Memory Set	Branch if $(Mn) = 1$	-	-	-	-	-	↕	DIR (b0)	00	dd rr	5
			-	-	-	-	-	↕	DIR (b1)	02	dd rr	5
			-	-	-	-	-	↕	DIR (b2)	04	dd rr	5
			-	-	-	-	-	↕	DIR (b3)	06	dd rr	5
			-	-	-	-	-	↕	DIR (b4)	08	dd rr	5
			-	-	-	-	-	↕	DIR (b5)	0A	dd rr	5
			-	-	-	-	-	↕	DIR (b6)	0C	dd rr	5
			-	-	-	-	-	↕	DIR (b7)	0E	dd rr	5
			-	-	-	-	-	-	DIR (b0)	10	dd	5
			-	-	-	-	-	-	DIR (b1)	12	dd	5
			-	-	-	-	-	-	DIR (b2)	14	dd	5

Table continues on the next page...

Table 10-3. Instruction Set Summary (continued)

Source Form	Operation	Description	Effect on CCR						Address Mode	Opcode	Operand	Bus Cycles
			V	H	I	N	Z	C				
BSET n,opr8a	Set Bit n in Memory	$M_n \leftarrow 1$	–	–	–	–	–	–	DIR (b3)	16	dd	5
			–	–	–	–	–	–	DIR (b4)	18	dd	5
			–	–	–	–	–	–	DIR (b5)	1A	dd	5
			–	–	–	–	–	–	DIR (b6)	1C	dd	5
			–	–	–	–	–	–	DIR (b7)	1E	dd	5
BSR rel	Branch to Subroutine	$PC \leftarrow (PC) + 0x0002$ push (PCL) $SP \leftarrow (SP) - 0x0001$ push (PCH) $SP \leftarrow (SP) - 0x0001$ $PC \leftarrow (PC) + rel$	–	–	–	–	–	–	REL	AD	rr	5
CBEQ opr8a,rel	Compare and Branch if Equal	Branch if (A) = (M)	–	–	–	–	–	–	DIR	31	dd rr	5
CBEQA #opr8i,rel		Branch if (A) = (M)	–	–	–	–	–	–	IMM	41	ii rr	4
CBEQX #opr8i,rel		Branch if (X) = (M)	–	–	–	–	–	–	IMM	51	ii rr	4
CBEQ oprx8,X+,rel		Branch if (A) = (M)	–	–	–	–	–	–	IX1+	61	ff rr	5
CBEQ ,X+,rel		Branch if (A) = (M)	–	–	–	–	–	–	IX+	71	rr	5
CBEQ oprx8,SP,rel		Branch if (A) = (M)	–	–	–	–	–	–	SP1	9E61	ff rr	6
CLC	Clear Carry Bit	$C \leftarrow 0$	–	–	–	–	–	0	INH	98		1
CLI	Clear Interrupt Mask Bit	$I \leftarrow 0$	–	–	0	–	–	–	INH	9A		1
CLR opr8a	Clear	$M \leftarrow 0x00$	0	–	–	0	1	–	DIR	3F	dd	5
CLRA		$A \leftarrow 0x00$	0	–	–	0	1	–	INH	4F		1
CLR X		$X \leftarrow 0x00$	0	–	–	0	1	–	INH	5F		1
CLR H		$H \leftarrow 0x00$	0	–	–	0	1	–	INH	8C		1
CLR oprx8,X		$M \leftarrow 0x00$	0	–	–	0	1	–	IX1	6F	ff	5
CLR ,X		$M \leftarrow 0x00$	0	–	–	0	1	–	IX	7F		4
CLR oprx8,SP		$M \leftarrow 0x00$	0	–	–	0	1	–	SP1	9E6F	ff	6
CMP #opr8i	Compare Accumulator with Memory	(A) – (M); (CCR Updated But Operands Not Changed)	↑	–	–	↑	↑	↑	IMM	A1	ii	2
CMP opr8a			↑	–	–	↑	↑	↑	DIR	B1	dd	3
CMP opr16a			↑	–	–	↑	↑	↑	EXT	C1	hh ll	4
CMP oprx16,X			↑	–	–	↑	↑	↑	IX2	D1	ee ff	4
CMP oprx8,X			↑	–	–	↑	↑	↑	IX1	E1	ff	3
CMP ,X			↑	–	–	↑	↑	↑	IX	F1		3
CMP oprx16,SP			↑	–	–	↑	↑	↑	SP2	9ED1	ee ff	5

Table continues on the next page...

Table 10-3. Instruction Set Summary (continued)

Source Form	Operation	Description	Effect on CCR						Address Mode	Opcode	Operand	Bus Cycles
			V	H	I	N	Z	C				
CMP oprx8,SP			↑	–	–	↑	↑	↑	SP1	9EE1	ff	4
COM opr8a	One's Complement	$M \leftarrow (M) = 0xFF - (M)$	0	–	–	↑	↑	1	DIR	33	dd	5
COMA		$A \leftarrow (A) = 0xFF - (A)$	0	–	–	↑	↑	1	INH	43		1
COMX		$X \leftarrow (X) = 0xFF - (X)$	0	–	–	↑	↑	1	INH	53		1
COM oprx8,X		$M \leftarrow (M) = 0xFF - (M)$	0	–	–	↑	↑	1	IX1	63	ff	5
COM ,X		$M \leftarrow (M) = 0xFF - (M)$	0	–	–	↑	↑	1	IX	73		4
COM oprx8,SP		$M \leftarrow (M) = 0xFF - (M)$	0	–	–	↑	↑	1	SP1	9E63	ff	6
CPHX opr16a	Compare Index Register (H:X) with Memory	(H:X) – (M:M + 0x0001); (CCR Updated But Operands Not Changed)	↑	–	–	↑	↑	↑	EXT	3E	hh ll	6
CPHX #opr16i			↑	–	–	↑	↑	↑	IMM	65	jj kk	3
CPHX opr8a			↑	–	–	↑	↑	↑	DIR	75	dd	5
CPHX oprx8,SP			↑	–	–	↑	↑	↑	SP1	9EF3	ff	6
CPX #opr8i			↑	–	–	↑	↑	↑	IMM	A3	ii	2
CPX opr8a	Compare X (Index Register Low) with Memory	(X) – (M); (CCR Updated But Operands Not Changed)	↑	–	–	↑	↑	↑	DIR	B3	dd	3
CPX opr16a			↑	–	–	↑	↑	↑	EXT	C3	hh ll	4
CPX oprx16,X			↑	–	–	↑	↑	↑	IX2	D3	ee ff	4
CPX oprx8,X			↑	–	–	↑	↑	↑	IX1	E3	ff	3
CPX ,X			↑	–	–	↑	↑	↑	IX	F3		3
CPX oprx16,SP			↑	–	–	↑	↑	↑	SP2	9ED3	ee ff	5
CPX oprx8,SP			↑	–	–	↑	↑	↑	SP1	9EE3	ff	4
DAA	Decimal Adjust Accumulator After ADD or ADC of BCD Values	(A) ₁₀	U	–	–	↑	↑	↑	INH	72		1
DBNZ opr8a,rel	Decrement and Branch if Not Zero	Decrement A, X, or M Branch if (result) ≠ 0 Affects X, Not H	–	–	–	–	–	–	DIR	3B	dd rr	7
DBNZA rel			–	–	–	–	–	–	INH	4B	rr	4
DBNZX rel			–	–	–	–	–	–	INH	5B	rr	4
DBNZ oprx8,X,rel			–	–	–	–	–	–	IX1	6B	ff rr	7
DBNZ ,X,rel			–	–	–	–	–	–	IX	7B	rr	6
DBNZ oprx8,SP,rel			–	–	–	–	–	–	SP1	9E6B	ff rr	8
DEC opr8a		$M \leftarrow (M) - 0x01$	↑	–	–	↑	↑	–	DIR	3A	dd	5
DECA		$A \leftarrow (A) - 0x01$	↑	–	–	↑	↑	–	INH	4A		1
DECX		$X \leftarrow (X) - 0x01$	↑	–	–	↑	↑	–	INH	5A		1

Table continues on the next page...

Table 10-3. Instruction Set Summary (continued)

Source Form	Operation	Description	Effect on CCR						Address Mode	Opcode	Operand	Bus Cycles
			V	H	I	N	Z	C				
DEC oprx8,X	Decrement	$M \leftarrow (M) - 0x01$	↓	–	–	↓	↓	–	IX1	6A	ff	5
DEC ,X		$M \leftarrow (M) - 0x01$	↓	–	–	↓	↓	–	IX	7A		4
DEC oprx8,SP		$M \leftarrow (M) - 0x01$	↓	–	–	↓	↓	–	SP1	9E6A	ff	6
DIV	Divide	$A \leftarrow (H:A) \div (X), H \leftarrow \text{Remainder}$	–	–	–	–	↓	↓	INH	52		6
EOR #opr8i	Exclusive OR Memory with Accumulator	$A \leftarrow (A \oplus M)$	0	–	–	↓	↓	–	IMM	A8	ii	2
EOR opr8a			0	–	–	↓	↓	–	DIR	B8	dd	3
EOR opr16a			0	–	–	↓	↓	–	EXT	C8	hh ll	4
EOR oprx16,X			0	–	–	↓	↓	–	IX2	D8	ee ff	4
EOR oprx8,X			0	–	–	↓	↓	–	IX1	E8	ff	3
EOR ,X			0	–	–	↓	↓	–	IX	F8		3
EOR oprx16,SP			0	–	–	↓	↓	–	SP2	9ED8	ee ff	5
EOR oprx8,SP			0	–	–	↓	↓	–	SP1	9EE8	ff	4
INC opr8a	Increment	$M \leftarrow (M) + 0x01$	↓	–	–	↓	↓	–	DIR	3C	dd	5
INCA		$A \leftarrow (A) + 0x01$	↓	–	–	↓	↓	–	INH	4C		1
INCX		$X \leftarrow (X) + 0x01$	↓	–	–	↓	↓	–	INH	5C		1
INC oprx8,X		$M \leftarrow (M) + 0x01$	↓	–	–	↓	↓	–	IX1	6C	ff	5
INC ,X		$M \leftarrow (M) + 0x01$	↓	–	–	↓	↓	–	IX	7C		4
INC oprx8,SP		$M \leftarrow (M) + 0x01$	↓	–	–	↓	↓	–	SP1	9E6C	ff	6
JMP opr8a	Jump	$PC \leftarrow \text{Jump Address}$							DIR	BC	dd	3
JMP opr16a									EXT	CC	hh ll	4
JMP oprx16,X			–	–	–	–	–	–	IX2	DC	ee ff	4
JMP oprx8,X									IX1	EC	ff	3
JMP ,X									IX	FC		3
JSR opr8a	Jump to Subroutine	$PC \leftarrow (PC) + n$ ($n = 1, 2,$ or 3) Push (PCL) $SP \leftarrow (SP) - 0x0001$ Push (PCH) $SP \leftarrow (SP) - 0x0001$ $PC \leftarrow \text{Unconditional Address}$	–	–	–	–	–	–	DIR	BD	dd	5
JSR opr16a			–	–	–	–	–	–	EXT	CD	hh ll	6
JSR oprx16,X			–	–	–	–	–	–	IX2	DD	ee ff	6
JSR oprx8,X			–	–	–	–	–	–	IX1	ED	ff	5
JSR ,X			–	–	–	–	–	–	IX	FD		5
LDA #opr8i	Load Accumulator from Memory	$A \leftarrow (M)$	0	–	–	↓	↓	–	IMM	A6	ii	2
LDA opr8a									DIR	B6	dd	3
LDA opr16a									EXT	C6	hh ll	4
LDA oprx16,X									IX2	D6	ee ff	4
LDA oprx8,X									IX1	E6	ff	3

Table continues on the next page...

Table 10-3. Instruction Set Summary (continued)

Source Form	Operation	Description	Effect on CCR						Address Mode	Opcode	Operand	Bus Cycles
			V	H	I	N	Z	C				
LDA ,X									IX	F6		3
LDA oprx16,SP									SP2	9ED6	ee ff	5
LDA oprx8,SP									SP1	9EE6	ff	4
LDHX #opr16i	Load Index Register (H:X) from Memory	$H:X \leftarrow (M:M + 0x0001)$	0	–	–	↑	↑	–	IMM	45	jj kk	3
LDHX opr8a			0	–	–	↑	↑	–	DIR	55	dd	4
LDHX opr16a			0	–	–	↑	↑	–	EXT	32	hh ll	5
LDHX ,X			0	–	–	↑	↑	–	IX	9EAE		5
LDHX oprx16,X			0	–	–	↑	↑	–	IX2	9EBE	ee ff	6
LDHX oprx8,X			0	–	–	↑	↑	–	IX1	9ECE	ff	5
LDHX oprx8,SP			0	–	–	↑	↑	–	SP1	9EFE	ff	5
LDX #opr8i	Load X (Index Register Low) from Memory	$X \leftarrow (M)$	0	–	–	↑	↑	–	IMM	AE	ii	2
LDX opr8a			0	–	–	↑	↑	–	DIR	BE	dd	3
LDX opr16a			0	–	–	↑	↑	–	EXT	CE	hh ll	4
LDX oprx16,X			0	–	–	↑	↑	–	IX2	DE	ee ff	4
LDX oprx8,X			0	–	–	↑	↑	–	IX1	EE	ff	3
LDX ,X			0	–	–	↑	↑	–	IX	FE		3
LDX oprx16,SP			0	–	–	↑	↑	–	SP2	9EDE	ee ff	5
LDX oprx8,SP			0	–	–	↑	↑	–	SP1	9EEE	ff	4
LSL opr8a	Logical Shift Left (Same as ASL)	$C \leftarrow \text{MSB}, \text{LSB} \leftarrow 0$	↑	–	–	↑	↑	↑	DIR	38	dd	5
LSLA			↑	–	–	↑	↑	↑	INH	48		1
LSLX			↑	–	–	↑	↑	↑	INH	58		1
LSL oprx8,X			↑	–	–	↑	↑	↑	IX1	68	ff	5
LSL ,X			↑	–	–	↑	↑	↑	IX	78		4
LSL oprx8,SP			↑	–	–	↑	↑	↑	SP1	9E68	ff	6
LSR opr8a	Logical Shift Right	$0 \rightarrow \text{MSB}, \text{LSB} \rightarrow C$	↑	–	–	0	↑	↑	DIR	34	dd	5
LSRA			↑	–	–	0	↑	↑	INH	44		1
LSRX			↑	–	–	0	↑	↑	INH	54		1
LSR oprx8,X			↑	–	–	0	↑	↑	IX1	64	ff	5
LSR ,X			↑	–	–	0	↑	↑	IX	74		4
LSR oprx8,SP			↑	–	–	0	↑	↑	SP1	9E64	ff	6
MOV opr8a,opr8a	Move	$(M)_{\text{destination}} \leftarrow (M)_{\text{source}}$	0	–	–	↑	↑	–	DIR/DIR	4E	dd	5
MOV opr8a,X+			0	–	–	↑	↑	–	DIR/IX+	5E	dd	5

Table continues on the next page...

Table 10-3. Instruction Set Summary (continued)

Source Form	Operation	Description	Effect on CCR						Address Mode	Opcode	Operand	Bus Cycles
			V	H	I	N	Z	C				
MOV #opr8i,opr8a		H:X $\leftarrow$ (H:X) + 0x0001 in IX+/DIR and DIR/IX+ Modes	0	–	–	$\uparrow$	$\uparrow$	–	IMM/DIR	6E	ii	4
MOV ,X+,opr8a			0	–	–	$\uparrow$	$\uparrow$	–	IX+/DIR	7E	dd	5
MUL	Unsigned multiply	X:A $\leftarrow$ (X) $\times$ (A)	–	0	–	–	–	0	INH	42		5
NEG opr8a	Negate (Two's Complement)	M $\leftarrow$ – (M) = 0x00 – (M)	$\uparrow$	–	–	$\uparrow$	$\uparrow$	$\uparrow$	DIR	30	dd	5
NEGA		A $\leftarrow$ – (A) = 0x00 – (A)	$\uparrow$	–	–	$\uparrow$	$\uparrow$	$\uparrow$	INH	40		1
NEGX		X $\leftarrow$ – (X) = 0x00 – (X)	$\uparrow$	–	–	$\uparrow$	$\uparrow$	$\uparrow$	INH	50		1
NEG oprx8,X		M $\leftarrow$ – (M) = 0x00 – (M)	$\uparrow$	–	–	$\uparrow$	$\uparrow$	$\uparrow$	IX1	60	ff	5
NEG ,X		M $\leftarrow$ – (M) = 0x00 – (M)	$\uparrow$	–	–	$\uparrow$	$\uparrow$	$\uparrow$	IX	70		4
NEG oprx8,SP		M $\leftarrow$ – (M) = 0x00 – (M)	$\uparrow$	–	–	$\uparrow$	$\uparrow$	$\uparrow$	SP1	9E60	ff	6
NOP	No Operation	Uses 1 Bus Cycle	–	–	–	–	–	–	INH	9D		1
NSA	Nibble Swap Accumulator	A $\leftarrow$ (A[3:0]:A[7:4])	–	–	–	–	–	–	INH	62		1
ORA #opr8i	Inclusive OR Accumulator and Memory	A $\leftarrow$ (A) (M)	0	–	–	$\uparrow$	$\uparrow$	–	IMM	AA	ii	2
ORA opr8a			0	–	–	$\uparrow$	$\uparrow$	–	DIR	BA	dd	3
ORA opr16a			0	–	–	$\uparrow$	$\uparrow$	–	EXT	CA	hh ll	4
ORA oprx16,X			0	–	–	$\uparrow$	$\uparrow$	–	IX2	DA	ee ff	4
ORA oprx8,X			0	–	–	$\uparrow$	$\uparrow$	–	IX1	EA	ff	3
ORA ,X			0	–	–	$\uparrow$	$\uparrow$	–	IX	FA		3
ORA oprx16,SP			0	–	–	$\uparrow$	$\uparrow$	–	SP2	9EDA	ee ff	5
ORA oprx8,SP			0	–	–	$\uparrow$	$\uparrow$	–	SP1	9EEA	ff	4
PSHA	Push Accumulator onto Stack	Push (A); SP $\leftarrow$ (SP) – 0x0001	–	–	–	–	–	–	INH	87		2
PSHH	Push H (Index Register High) onto Stack	Push (H); SP $\leftarrow$ (SP) – 0x0001	–	–	–	–	–	–	INH	8B		2
PSHX	Push X (Index Register Low) onto Stack	Push (X); SP $\leftarrow$ (SP) – 0x0001	–	–	–	–	–	–	INH	89		2
PULA	Pull Accumulator from Stack	SP $\leftarrow$ (SP + 0x0001); Pull (A)	–	–	–	–	–	–	INH	86		3
PULH	Pull H (Index Register High) from Stack	SP $\leftarrow$ (SP + 0x0001); Pull (H)	–	–	–	–	–	–	INH	8A		3
PULX	Pull X (Index Register Low) from Stack	SP $\leftarrow$ (SP + 0x0001); Pull (X)	–	–	–	–	–	–	INH	88		3
ROL opr8a			$\uparrow$	–	–	$\uparrow$	$\uparrow$	$\uparrow$	DIR	39	dd	5
ROLA			$\uparrow$	–	–	$\uparrow$	$\uparrow$	$\uparrow$	INH	49		1

Table continues on the next page...

Table 10-3. Instruction Set Summary (continued)

Source Form	Operation	Description	Effect on CCR						Address Mode	Opcode	Operand	Bus Cycles
			V	H	I	N	Z	C				
ROLX	Rotate Left through Carry	$C \leftarrow \text{MSB}, \text{LSB} \leftarrow C$	$\uparrow$	–	–	$\uparrow$	$\uparrow$	$\uparrow$	INH	59		1
ROL oprx8,X			$\uparrow$	–	–	$\uparrow$	$\uparrow$	$\uparrow$	IX1	69	ff	5
ROL ,X			$\uparrow$	–	–	$\uparrow$	$\uparrow$	$\uparrow$	IX	79		4
ROL oprx8,SP			$\uparrow$	–	–	$\uparrow$	$\uparrow$	$\uparrow$	SP1	9E69	ff	6
ROR opr8a	Rotate Right through Carry	$\text{LSB} \rightarrow C, C \rightarrow \text{MSB}$	$\uparrow$	–	–	$\uparrow$	$\uparrow$	$\uparrow$	DIR	36	dd	5
RORA			$\uparrow$	–	–	$\uparrow$	$\uparrow$	$\uparrow$	INH	46		1
RORX			$\uparrow$	–	–	$\uparrow$	$\uparrow$	$\uparrow$	INH	56		1
ROR oprx8,X			$\uparrow$	–	–	$\uparrow$	$\uparrow$	$\uparrow$	IX1	66	ff	5
ROR ,X			$\uparrow$	–	–	$\uparrow$	$\uparrow$	$\uparrow$	IX	76		4
ROR oprx8,SP			$\uparrow$	–	–	$\uparrow$	$\uparrow$	$\uparrow$	SP1	9E66	ff	6
RSP	Reset Stack Pointer	$\text{SP} \leftarrow 0\text{xFF}$ (High Byte Not Affected)	–	–	–	–	–	–	INH	9C		1
RTI	Return from Interrupt	$\text{SP} \leftarrow (\text{SP}) + 0\text{x0001}$, Pull (CCR) $\text{SP} \leftarrow (\text{SP}) + 0\text{x0001}$, Pull (A) $\text{SP} \leftarrow (\text{SP}) + 0\text{x0001}$, Pull (X) $\text{SP} \leftarrow (\text{SP}) + 0\text{x0001}$, Pull (PCH) $\text{SP} \leftarrow (\text{SP}) + 0\text{x0001}$, Pull (PCL)	$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$	INH	80		9
RTS	Return from Subroutine	$\text{SP} \leftarrow \text{SP} + 0\text{x0001}$, Pull (PCH) $\text{SP} \leftarrow \text{SP} + 0\text{x0001}$, Pull (PCL)	–	–	–	–	–	–	INH	81		6
SBC #opr8i	Subtract with Carry	$A \leftarrow (A) - (M) - (C)$	$\uparrow$	–	–	$\uparrow$	$\uparrow$	$\uparrow$	IMM	A2	ii	2
SBC opr8a			$\uparrow$	–	–	$\uparrow$	$\uparrow$	$\uparrow$	DIR	B2	dd	3
SBC opr16a			$\uparrow$	–	–	$\uparrow$	$\uparrow$	$\uparrow$	EXT	C2	hh ll	4
SBC oprx16,X			$\uparrow$	–	–	$\uparrow$	$\uparrow$	$\uparrow$	IX2	D2	ee ff	4
SBC oprx8,X			$\uparrow$	–	–	$\uparrow$	$\uparrow$	$\uparrow$	IX1	E2	ff	3
SBC ,X			$\uparrow$	–	–	$\uparrow$	$\uparrow$	$\uparrow$	IX	F2		3
SBC oprx16,SP			$\uparrow$	–	–	$\uparrow$	$\uparrow$	$\uparrow$	SP2	9ED2	ee ff	5
SBC oprx8,SP			$\uparrow$	–	–	$\uparrow$	$\uparrow$	$\uparrow$	SP1	9EE2	ff	4
SEC	Set Carry Bit	$C \leftarrow 1$	–	–	–	–	–	1	INH	99		1
SEI	Set Interrupt Mask Bit	$I \leftarrow 1$	–	–	1	–	–	–	INH	9B		1

Table continues on the next page...

Table 10-3. Instruction Set Summary (continued)

Source Form	Operation	Description	Effect on CCR						Address Mode	Opcode	Operand	Bus Cycles
			V	H	I	N	Z	C				
STA opr8a	Store Accumulator in Memory	$M \leftarrow (A)$	0	–	–	↑	↑	–	DIR	B7	dd	3
STA opr16a			0	–	–	↑	↑	–	EXT	C7	hh ll	4
STA oprx16,X			0	–	–	↑	↑	–	IX2	D7	ee ff	4
STA oprx8,X			0	–	–	↑	↑	–	IX1	E7	ff	3
STA ,X			0	–	–	↑	↑	–	IX	F7		2
STA oprx16,SP			0	–	–	↑	↑	–	SP2	9ED7	ee ff	5
STA oprx8,SP			0	–	–	↑	↑	–	SP1	9EE7	ff	4
STHX opr8a	Store H:X (Index Reg.)	$(M:M + 0x0001) \leftarrow (H:X)$	0	–	–	↑	↑	–	DIR	35	dd	4
STHX opr16a			0	–	–	↑	↑	–	EXT	96	hh ll	5
STHX oprx8,SP			0	–	–	↑	↑	–	SP1	9EFF	ff	5
STOP	Enable Interrupts: Stop Processing. Refer to MCU Documentation.	I bit $\leftarrow$ 0; Stop Processing	–	–	0	–	–	–	INH	8E		3+
STX opr8a	Store X (Low 8 Bits of Index Register) in Memory	$M \leftarrow (X)$	0	–	–	↑	↑	–	DIR	BF	dd	3
STX opr16a			0	–	–	↑	↑	–	EXT	CF	hh ll	4
STX oprx16,X			0	–	–	↑	↑	–	IX2	DF	ee ff	4
STX oprx8,X			0	–	–	↑	↑	–	IX1	EF	ff	3
STX ,X			0	–	–	↑	↑	–	IX	FF		2
STX oprx16,SP			0	–	–	↑	↑	–	SP2	9EDF	ee ff	5
STX oprx8,SP			0	–	–	↑	↑	–	SP1	9EEF	ff	4
SUB #opr8i	Subtract	$A \leftarrow (A) - (M)$	↑	–	–	↑	↑	↑	IMM	A0	ii	2
SUB opr8a			↑	–	–	↑	↑	↑	DIR	B0	dd	3
SUB opr16a			↑	–	–	↑	↑	↑	EXT	C0	hh ll	4
SUB oprx16,X			↑	–	–	↑	↑	↑	IX2	D0	ee ff	4
SUB oprx8,X			↑	–	–	↑	↑	↑	IX1	E0	ff	3
SUB ,X			↑	–	–	↑	↑	↑	IX	F0		3
SUB oprx16,SP			↑	–	–	↑	↑	↑	SP2	9ED0	ee ff	5
SUB oprx8,SP			↑	–	–	↑	↑	↑	SP1	9EE0	ff	4
		PC $\leftarrow$ (PC) + 0x0001 Push (PCL) SP $\leftarrow$ (SP) – 0x0001 Push (PCH) SP $\leftarrow$ (SP) – 0x0001, Push (X) SP $\leftarrow$ (SP) – 0x0001 Push (A)										

Table continues on the next page...

Table 10-3. Instruction Set Summary (continued)

Source Form	Operation	Description	Effect on CCR						Address Mode	Opcode	Operand	Bus Cycles
			V	H	I	N	Z	C				
SWI	Software Interrupt	$SP \leftarrow (SP) - 0x0001$ Push (CCR) $SP \leftarrow (SP) - 0x0001$ $I \leftarrow 1$ $PCH \leftarrow$ Interrupt Vector High Byte $PCL \leftarrow$ Interrupt Vector Low Byte	-	-	1	-	-	-	INH	83		11
TAP	Transfer Accumulator to CCR	$CCR \leftarrow (A)$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	INH	84		1
TAX	Transfer Accumulator to X (Index Register Low)	$X \leftarrow (A)$	-	-	-	-	-	-	INH	97		1
TPA	Transfer CCR to Accumulator	$A \leftarrow (CCR)$	-	-	-	-	-	-	INH	85		1
TST opr8a	Test for Negative or Zero	$(M) - 0x00$	0	-	-	$\updownarrow$	$\updownarrow$	-	DIR	3D	dd	4
TSTA		$(A) - 0x00$	0	-	-	$\updownarrow$	$\updownarrow$	-	INH	4D		1
TSTX		$(X) - 0x00$	0	-	-	$\updownarrow$	$\updownarrow$	-	INH	5D		1
TST oprx8,X		$(M) - 0x00$	0	-	-	$\updownarrow$	$\updownarrow$	-	IX1	6D	ff	4
TST ,X		$(M) - 0x00$	0	-	-	$\updownarrow$	$\updownarrow$	-	IX	7D		3
TST oprx8,SP		$(M) - 0x00$	0	-	-	$\updownarrow$	$\updownarrow$	-	SP1	9E6D	ff	5
TSX	Transfer SP to Index Register	$H:X \leftarrow (SP) + 0x0001$	-	-	-	-	-	-	INH	95		2
TXA	Transfer X (Index Reg. Low) to Accumulator	$A \leftarrow (X)$	-	-	-	-	-	-	INH	9F		1
TXS	Transfer Index Register to SP	$SP \leftarrow (H:X) - 0x0001$	-	-	-	-	-	-	INH	94		2
WAIT	Enable Interrupts Wait for Interrupt	$I \text{ bit} \leftarrow 0$, Halt CPU	-	-	0	-	-	-	INH	8F		3+